3.4.2 Regularized Linear Models: ElasticNet, LASSO, & Ridge

High dimensional cross-sectional asset pricing settings with P characteristic features suffer from multicollinearity and sample overfitting under standard OLS. Regularized linear models introduce penalty structures to constrain coefficient magnitudes and perform feature selection (Kozak et al., 2020).

ElasticNet Objective Function

Let $\mathbf{y} \in \mathbb{R}^N$ denote the vector of standardized forward returns across all asset-day observations, and let $\mathbf{X} \in \mathbb{R}^{N \times P}$ denote the matrix of standardized stock characteristics. The ElasticNet estimator (Zou and Hastie, 2005) solves the convex optimization problem:

$$\min_{\boldsymbol{\beta} \in \mathbb{R}^P} \mathcal{L}_{\mathrm{EN}}(\boldsymbol{\beta}) = \frac{1}{2N} \|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|_2^2 + \lambda \left[\alpha \|\boldsymbol{\beta}\|_1 + \frac{1-\alpha}{2} \|\boldsymbol{\beta}\|_2^2 \right] \tag{3.18}$$

where $\lambda > 0$ controls the penalty intensity, $\alpha \in [0, 1]$ governs the ratio between the L_1 norm $\|\boldsymbol{\beta}\|_1 = \sum_{j=1}^{P} |\beta_j|$ and the squared L_2 norm $\|\boldsymbol{\beta}\|_2^2 = \sum_{j=1}^{P} \beta_j^2$. Setting $\alpha = 1.0$ yields the LASSO estimator (Tibshirani, 1996), while $\alpha = 0.0$ yields Ridge regression (Hoerl and Kennard, 1970).

Subgradient Calculus & Characterization of LASSO (L_1) Sparsity

Because the L_1 norm is non-differentiable at $\beta_j = 0$, optimization requires subgradient calculus. The subdifferential $\partial|\beta_j|$ of the absolute value function is defined as:

$$\partial|\beta_j| = \begin{cases} \{+1\} & \text{if } \beta_j > 0 \\ \{-1\} & \text{if } \beta_j < 0 \\ [-1, 1] & \text{if } \beta_j = 0 \end{cases} \tag{3.19}$$

To derive the coordinate descent update rule for feature j, assume that the columns of $\mathbf{X}$ are standardized such that $\mathbf{x}_j^T \mathbf{x}_j = N$. Isolate feature j by defining the partial residual vector $\mathbf{r}^{(j)} = \mathbf{y} - \sum_{k \neq j} \mathbf{x}_k \beta_k$. The unregularized OLS update for β_j given fixed $\boldsymbol{\beta}_{-j}$ is $\hat{\beta}_j^{\text{OLS}} = \frac{1}{N} \mathbf{x}_j^T \mathbf{r}^{(j)}$.

The first order optimality condition for the ElasticNet objective requires $0 \in \partial_{\beta_j} \mathcal{L}_{\text{EN}}(\boldsymbol{\beta})$:

$$-\frac{1}{N} \mathbf{x}_j^T (\mathbf{y} - \mathbf{X}\boldsymbol{\beta}) + \lambda(1-\alpha)\beta_j + \lambda\alpha v_j = 0, \quad \text{where } v_j \in \partial|\beta_j| \tag{3.20}$$

Substituting $\mathbf{y} - \mathbf{X}\boldsymbol{\beta} = \mathbf{r}^{(j)} - \mathbf{x}_j \beta_j$ and using $\frac{1}{N} \mathbf{x}_j^T \mathbf{x}_j = 1$:

$$-\hat{\beta}_j^{\text{OLS}} + \beta_j + \lambda(1-\alpha)\beta_j + \lambda\alpha v_j = 0 \implies \beta_j[1 + \lambda(1-\alpha)] + \lambda\alpha v_j = \hat{\beta}_j^{\text{OLS}} \tag{3.21}$$

Solving Equation (3.21) under three case conditions for $\hat{\beta}_j^{\text{OLS}}$:

1. **Case 1:** $\beta_j > 0$. Here $v_j = +1$. Equation (3.21) gives $\beta_j[1 + \lambda(1-\alpha)] = \hat{\beta}_j^{\text{OLS}} - \lambda\alpha$, requiring $\hat{\beta}_j^{\text{OLS}} > \lambda\alpha$. Thus:

$$\beta_j^* = \frac{\hat{\beta}_j^{\text{OLS}} - \lambda\alpha}{1 + \lambda(1-\alpha)}$$

2. **Case 2:** $\beta_j < 0$. Here $v_j = -1$. Equation (3.21) gives $\beta_j[1 + \lambda(1-\alpha)] = \hat{\beta}_j^{\text{OLS}} + \lambda\alpha$, requiring $\hat{\beta}_j^{\text{OLS}} < -\lambda\alpha$. Thus:

$$\beta_j^* = \frac{\hat{\beta}_j^{\text{OLS}} + \lambda\alpha}{1 + \lambda(1-\alpha)}$$

3. **Case 3:** $\beta_j = 0$. Here $v_j \in [-1, 1]$. Equation (3.21) requires $|\hat{\beta}_j^{\text{OLS}}| \leq \lambda\alpha$. Thus, $\beta_j^* = 0$.

Combining these cases yields the closed-form Soft-Thresholding Operator $\mathcal{S}_\kappa(z) = \text{sign}(z)\max(0, |z| - \kappa)$:

$$\hat{\beta}_j^{\text{EN}} = \frac{\mathcal{S}_{\lambda\alpha}\left(\hat{\beta}_j^{\text{OLS}}\right)}{1 + \lambda(1-\alpha)} \tag{3.22}$$

Setting $\alpha = 1.0$ yields the standard LASSO soft-thresholded estimator $\hat{\beta}_j^{\text{LASSO}} = \mathcal{S}_\lambda\left(\hat{\beta}_j^{\text{OLS}}\right)$, which sets coefficients to zero whenever $|\hat{\beta}_j^{\text{OLS}}| \leq \lambda$, performing feature selection.

Ridge (L_2) Shrinkage Bounds & SVD Derivation

For Ridge regression ($\alpha = 0$), the objective function is smooth and differentiable. Differentiating $\mathcal{L}_{\text{Ridge}}(\boldsymbol{\beta})$ with respect to $\boldsymbol{\beta}$ yields the closed-form Ridge estimator:

$$\hat{\boldsymbol{\beta}}_{\text{Ridge}} = \left(\mathbf{X}^T\mathbf{X} + N\lambda\mathbf{I}_P\right)^{-1}\mathbf{X}^T\mathbf{y} \tag{3.23}$$

To derive L_2 shrinkage bounds, consider the Singular Value Decomposition (SVD) of the centered design matrix $\mathbf{X} = \mathbf{U}\boldsymbol{\Sigma}\mathbf{V}^T$, where $\mathbf{U} \in \mathbb{R}^{N\times P}$ and $\mathbf{V} \in \mathbb{R}^{P\times P}$ are orthonormal matrices ($\mathbf{U}^T\mathbf{U} = \mathbf{I}_P$, $\mathbf{V}^T\mathbf{V} = \mathbf{I}_P$), and $\boldsymbol{\Sigma} = \text{diag}(\sigma_1, \sigma_2, \ldots, \sigma_P)$ contains the singular values $\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_P > 0$.

Substituting the SVD into the standard OLS estimator $\hat{\boldsymbol{\beta}}_{\text{OLS}} = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{y}$:

$$\hat{\boldsymbol{\beta}}_{\text{OLS}} = \left(\mathbf{V}\boldsymbol{\Sigma}^2\mathbf{V}^T\right)^{-1}\mathbf{V}\boldsymbol{\Sigma}\mathbf{U}^T\mathbf{y} = \mathbf{V}\boldsymbol{\Sigma}^{-1}\mathbf{U}^T\mathbf{y} \tag{3.24}$$

Substituting the SVD into the Ridge estimator Equation (3.23):

$$\begin{aligned}
\hat{\boldsymbol{\beta}}_{\text{Ridge}} &= \left(\mathbf{V}\boldsymbol{\Sigma}^2\mathbf{V}^T + N\lambda\mathbf{V}\mathbf{I}_P\mathbf{V}^T\right)^{-1}\mathbf{V}\boldsymbol{\Sigma}\mathbf{U}^T\mathbf{y} && (3.25)\\
&= \left(\mathbf{V}[\boldsymbol{\Sigma}^2 + N\lambda\mathbf{I}_P]\mathbf{V}^T\right)^{-1}\mathbf{V}\boldsymbol{\Sigma}\mathbf{U}^T\mathbf{y} && (3.26)\\
&= \mathbf{V}\left(\boldsymbol{\Sigma}^2 + N\lambda\mathbf{I}_P\right)^{-1}\boldsymbol{\Sigma}\mathbf{U}^T\mathbf{y} && (3.27)
\end{aligned}$$

Expressing $\hat{\boldsymbol{\beta}}_{\text{Ridge}}$ directly as a transformation of the unregularized OLS estimator $\hat{\boldsymbol{\beta}}_{\text{OLS}}$:

$$\begin{aligned}
\hat{\boldsymbol{\beta}}_{\text{Ridge}} &= \mathbf{V}\left(\boldsymbol{\Sigma}^2 + N\lambda\mathbf{I}_P\right)^{-1}\boldsymbol{\Sigma}^2\boldsymbol{\Sigma}^{-1}\mathbf{U}^T\mathbf{y} && (3.28)\\
&= \mathbf{V}\,\text{diag}\left(\frac{\sigma_j^2}{\sigma_j^2 + N\lambda}\right)\mathbf{V}^T\hat{\boldsymbol{\beta}}_{\text{OLS}} && (3.29)
\end{aligned}$$

Because $N\lambda > 0$ and $\sigma_j^2 > 0$, the shrinkage factor satisfies:

$$0 < \frac{\sigma_j^2}{\sigma_j^2 + N\lambda} < 1, \quad \forall j \in \{1, \ldots, P\} \tag{3.30}$$

This shows that Ridge regression shrinks the OLS parameter vector along the principal component directions of $\mathbf{X}$, with greater shrinkage applied to components with smaller singular values σ_j^2. Consequently, $\|\hat{\boldsymbol{\beta}}_{\text{Ridge}}\|_2 < \|\hat{\boldsymbol{\beta}}_{\text{OLS}}\|_2$ for all $\lambda > 0$, bounding parameter variance and stabilizing prediction under collinearity.

3.4.3 Non-Linear Tree Ensembles: Random Forest & XGBoost

Decision tree ensembles map high dimensional characteristic spaces into predicted returns by partitioning features into hyper-rectangular regions (Gu et al., 2020).

Random Forest Bagging Variance Reduction Proof

The Random Forest regressor (Breiman, 2001) constructs an ensemble of M decorrelated decision trees $\{T_1(\mathbf{x}), T_2(\mathbf{x}), \ldots, T_M(\mathbf{x})\}$, each trained on a bootstrap sample of size N with randomized feature subspace selection (m_{try}). While the classic Breiman (2001) rule of thumb sets $m_{\text{try}} = \lfloor\sqrt{P}\rfloor = \lfloor\sqrt{53}\rfloor = 7$ (a feature subsample ratio of $7/53 \approx 0.13$), cross-validation tuning across expanding folds yields an optimal ratio of $m_{\text{try}}/P = 0.40$ ($m_{\text{try}} = 21$ features per node split), which prevents under-fitting when features are correlated within modalities. The ensemble prediction is the average:

$$\hat{f}_{\text{RF}}(\mathbf{x}) = \frac{1}{M}\sum_{m=1}^{M} T_m(\mathbf{x}) \tag{3.31}$$

Assume each individual tree $T_m(\mathbf{x})$ is identically distributed with variance $\text{Var}(T_m(\mathbf{x})) = \sigma^2$, and that any pair of distinct trees (T_m, T_l) for $m \neq l$ has pairwise correlation $\text{Corr}(T_m(\mathbf{x}), T_l(\mathbf{x})) = \rho$.

Proof of Variance Reduction Dynamics. Compute the variance of the ensemble estimator $\hat{f}_{\text{RF}}(\mathbf{x})$:

$$\operatorname{Var}\left(\hat{f}_{\text{RF}}(\mathbf{x})\right) = \operatorname{Var}\left(\frac{1}{M}\sum_{m=1}^{M} T_m(\mathbf{x})\right) \tag{3.32}$$

$$= \frac{1}{M^2}\left[\sum_{m=1}^{M} \operatorname{Var}(T_m(\mathbf{x})) + \sum_{m=1}^{M}\sum_{l\neq m}^{M} \operatorname{Cov}(T_m(\mathbf{x}), T_l(\mathbf{x}))\right] \tag{3.33}$$

Substitute $\operatorname{Var}(T_m(\mathbf{x})) = \sigma^2$ and $\operatorname{Cov}(T_m(\mathbf{x}), T_l(\mathbf{x})) = \rho\sigma^2$:

$$\operatorname{Var}\left(\hat{f}_{\text{RF}}(\mathbf{x})\right) = \frac{1}{M^2}\left[M\sigma^2 + M(M-1)\rho\sigma^2\right] \tag{3.34}$$

$$= \frac{\sigma^2}{M} + \frac{M-1}{M}\rho\sigma^2 = \rho\sigma^2 + \frac{1-\rho}{M}\sigma^2 \tag{3.35}$$

□

Taking the limit as the number of trees $M \to \infty$:

$$\lim_{M\to\infty} \operatorname{Var}\left(\hat{f}_{\text{RF}}(\mathbf{x})\right) = \rho\sigma^2 \tag{3.36}$$

Equation (3.35) demonstrates the variance-reduction mechanism of Random Forest:

1. **Bagging** (bootstrap aggregation) reduces the term $\frac{1-\rho}{M}\sigma^2$ toward zero as the ensemble size M increases.

2. **Random Feature Subspace Selection** reduces inter-tree correlation ρ, lowering the variance floor $\rho\sigma^2$.

XGBoost Second-Order Taylor Expansion Loss Optimization

Extreme Gradient Boosting (XGBoost) (Chen and Guestrin, 2016) builds an additive model of M decision trees sequentially: $\hat{y}_i^{(m)} = \hat{y}_i^{(m-1)} + f_m(\mathbf{x}_i)$. At step m, the regularized objective function to minimize is:

$$\mathcal{L}^{(m)} = \sum_{i=1}^{N} l\left(y_i, \hat{y}_i^{(m-1)} + f_m(\mathbf{x}_i)\right) + \Omega(f_m) \tag{3.37}$$

where $l(\cdot, \cdot)$ is a differentiable convex loss function, and the tree complexity regularization term $\Omega(f_m)$ is defined for a tree with T leaf nodes and leaf weight vector $\mathbf{w} = [w_1, \ldots, w_T]^T \in \mathbb{R}^T$ as:

$$\Omega(f_m) = \gamma T + \frac{1}{2}\lambda\|\mathbf{w}\|_2^2 = \gamma T + \frac{1}{2}\lambda \sum_{j=1}^{T} w_j^2 \tag{3.38}$$

Applying a second order Taylor expansion to the loss function around the previous iteration forecast $\hat{y}_i^{(m-1)}$:

$$l\left(y_i, \hat{y}_i^{(m-1)} + f_m(\mathbf{x}_i)\right) \approx l\left(y_i, \hat{y}_i^{(m-1)}\right) + g_i f_m(\mathbf{x}_i) + \frac{1}{2}h_i f_m(\mathbf{x}_i)^2 \tag{3.39}$$

where the first order gradient g_i and second order Hessian h_i are defined as:

$$g_i = \frac{\partial l\left(y_i, \hat{y}_i^{(m-1)}\right)}{\partial \hat{y}_i^{(m-1)}}, \quad h_i = \frac{\partial^2 l\left(y_i, \hat{y}_i^{(m-1)}\right)}{\partial \left(\hat{y}_i^{(m-1)}\right)^2} \tag{3.40}$$

Removing constant terms $l(y_i, \hat{y}_i^{(m-1)})$ independent of tree structure f_m, the simplified objective at step m becomes:

$$\tilde{\mathcal{L}}^{(m)} = \sum_{i=1}^{N} \left[g_i f_m(\mathbf{x}_i) + \frac{1}{2}h_i f_m(\mathbf{x}_i)^2\right] + \gamma T + \frac{1}{2}\lambda \sum_{j=1}^{T} w_j^2 \tag{3.41}$$

Let $I_j = \{i \mid q(\mathbf{x}_i) = j\}$ represent the instance set mapped to leaf node j by structure mapping function $q : \mathbb{R}^P \to \{1, \ldots, T\}$. Rewriting the summation across individual samples i as a summation over leaf nodes j:

$$\tilde{\mathcal{L}}^{(m)} = \sum_{j=1}^{T} \left[\left(\sum_{i \in I_j} g_i \right) w_j + \frac{1}{2} \left(\sum_{i \in I_j} h_i + \lambda \right) w_j^2 \right] + \gamma T \tag{3.42}$$

Define aggregate leaf gradient $G_j = \sum_{i \in I_j} g_i$ and aggregate leaf Hessian $H_j = \sum_{i \in I_j} h_i$:

$$\tilde{\mathcal{L}}^{(m)} = \sum_{j=1}^{T} \left[G_j w_j + \frac{1}{2}(H_j + \lambda) w_j^2 \right] + \gamma T \tag{3.43}$$

Taking the partial derivative of $\tilde{\mathcal{L}}^{(m)}$ with respect to weight w_j and setting it to zero yields the optimal weight w_j^* for leaf j:

$$\frac{\partial \tilde{\mathcal{L}}^{(m)}}{\partial w_j} = G_j + (H_j + \lambda) w_j = 0 \implies w_j^* = -\frac{G_j}{H_j + \lambda} \tag{3.44}$$

Substituting w_j^* back into the objective yields the minimized loss value (tree structure quality score):

$$\tilde{\mathcal{L}}^{(m)*} = -\frac{1}{2} \sum_{j=1}^{T} \frac{G_j^2}{H_j + \lambda} + \gamma T \tag{3.45}$$

When evaluating candidate node splits during tree construction, the gain score for splitting a node into left instance subset I_L and right instance subset I_R is evaluated as:

$$\text{Gain} = \frac{1}{2} \left[\frac{G_L^2}{H_L + \lambda} + \frac{G_R^2}{H_R + \lambda} - \frac{(G_L + G_R)^2}{H_L + H_R + \lambda} \right] - \gamma \tag{3.46}$$

where γ acts as a penalty threshold requiring the improvement in loss reduction to exceed γ before authorizing a node split.

3.4.4 Baseline Sequence Encoder: PyTorch LSTM

To encode temporal patterns across a sequential lookback window of $T = 20$ trading days, I use a multilayer Long Short-Term Memory (LSTM) network (Hochreiter and Schmidhuber, 1997).

For input vector $\mathbf{x}_t \in \mathbb{R}^{d_{\text{in}}}$ at sequence step $t \in \{1, \ldots, T\}$, previous hidden state $\mathbf{h}_{t-1} \in \mathbb{R}^D$, and previous cell state $\mathbf{c}_{t-1} \in \mathbb{R}^D$, the PyTorch LSTM cell equations are formulated as:

$$\mathbf{i}_t = \sigma\left(\mathbf{W}_{xi}\mathbf{x}_t + \mathbf{W}_{hi}\mathbf{h}_{t-1} + \mathbf{b}_{ii} + \mathbf{b}_{hi}\right) \quad \text{(Input Gate)} \tag{3.47}$$

$$\mathbf{f}_t = \sigma\left(\mathbf{W}_{xf}\mathbf{x}_t + \mathbf{W}_{hf}\mathbf{h}_{t-1} + \mathbf{b}_{if} + \mathbf{b}_{hf}\right) \quad \text{(Forget Gate)} \tag{3.48}$$

$$\mathbf{o}_t = \sigma\left(\mathbf{W}_{xo}\mathbf{x}_t + \mathbf{W}_{ho}\mathbf{h}_{t-1} + \mathbf{b}_{io} + \mathbf{b}_{ho}\right) \quad \text{(Output Gate)} \tag{3.49}$$

$$\tilde{\mathbf{c}}_t = \tanh\left(\mathbf{W}_{xc}\mathbf{x}_t + \mathbf{W}_{hc}\mathbf{h}_{t-1} + \mathbf{b}_{ic} + \mathbf{b}_{hc}\right) \quad \text{(Cell Candidate)} \tag{3.50}$$

$$\mathbf{c}_t = \mathbf{f}_t \odot \mathbf{c}_{t-1} + \mathbf{i}_t \odot \tilde{\mathbf{c}}_t \quad \text{(Cell State Update)} \tag{3.51}$$

$$\mathbf{h}_t = \mathbf{o}_t \odot \tanh(\mathbf{c}_t) \quad \text{(Hidden State Output)} \tag{3.52}$$

where $\sigma(z) = (1 + e^{-z})^{-1}$ is the element-wise logistic sigmoid function, $\tanh(z) = \frac{e^z - e^{-z}}{e^z + e^{-z}}$ is the hyperbolic tangent activation, and $\odot$ represents the Hadamard (element-wise) product. Weight matrices $\mathbf{W}_{x\cdot} \in \mathbb{R}^{D \times d_{\text{in}}}$, $\mathbf{W}_{h\cdot} \in \mathbb{R}^{D \times D}$, and bias vectors $\mathbf{b}_{\cdot} \in \mathbb{R}^D$ represent trainable parameters.

The linear cell state update $\mathbf{c}_t = \mathbf{f}_t \odot \mathbf{c}_{t-1} + \mathbf{i}_t \odot \tilde{\mathbf{c}}_t$ provides a constant error carousel. The partial Jacobian $\frac{\partial \mathbf{c}_t}{\partial \mathbf{c}_{t-1}} = \operatorname{diag}(\mathbf{f}_t)$ helps mitigate vanishing gradients over temporal sequences when the forget gate is active ($\mathbf{f}_t \approx \mathbf{1}$). The final hidden state $\mathbf{h}_T$ summarizes the sequence and is mapped via a linear projection head to yield predictions $\hat{y}_{i,t+1} = \mathbf{w}_y^T \mathbf{h}_T + b_y$.

3.5 Custom Architecture: TFDMGA Network

The **Temporal Fusion Deep Multimodal Gated Attention (TFDMGA)** network is designed to address three limitations of standard models in asset pricing: (1) feature scale and frequency differences across modalities, (2) inter-modal characteristic dependencies, and (3) macroeconomic regime sensitivity.

Figure 3.5 shows the complete neural network architecture of the TFDMGA model.

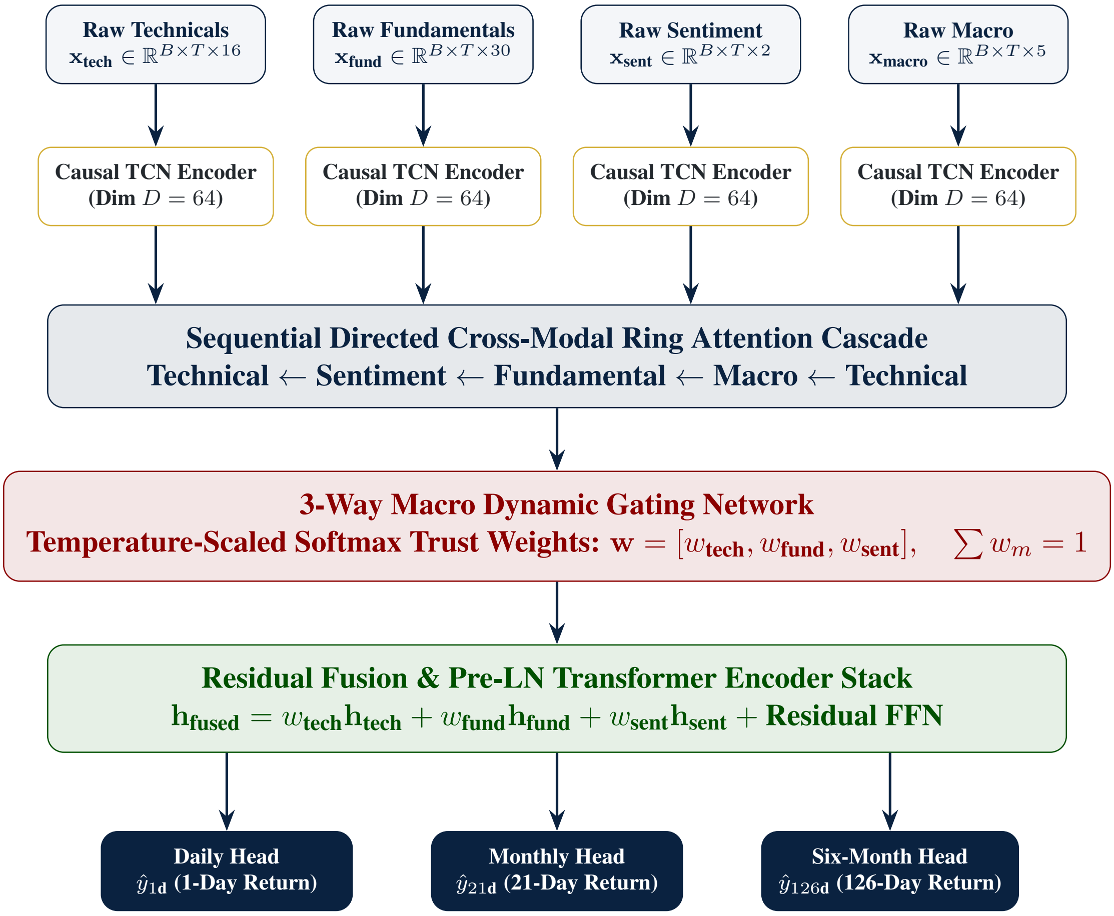

Figure 3.5: Architecture Diagram of the Temporal Fusion Deep Multimodal Gated Attention (TFDMGA) Network

3.5.1 Temporal Feature Encoding via Causal 1D Dilated TCN

Financial input features are partitioned into four distinct modalities: Technical Signals ($\mathbf{x}_{\text{tech}} \in \mathbb{R}^{B\times T\times 16}$), Fundamental Metrics ($\mathbf{x}_{\text{fund}} \in \mathbb{R}^{B\times T\times 30}$), Sentiment Scores ($\mathbf{x}_{\text{sent}} \in \mathbb{R}^{B\times T\times 2}$), and Macroeconomic Series ($\mathbf{x}_{\text{macro}} \in \mathbb{R}^{B\times T\times 5}$).

Each modality sequence is processed by a Causal 1D Dilated Temporal Convolutional Network (TCN) encoder (Bai et al., 2018).

Let $\mathbf{x}_m(t) \in \mathbb{R}^{C_{\text{in}}}$ denote the input vector for modality m at sequence step $t \in \{1, \ldots, T\}$. A 1D dilated convolution with kernel filter $f : \{0, \ldots, K-1\} \to \mathbb{R}^{C_{\text{out}}\times C_{\text{in}}}$ and dilation factor d is formulated as:

$$(y *_d f)(t) = \sum_{k=0}^{K-1} f(k) \cdot \mathbf{x}_m(t - d \cdot k) \tag{3.53}$$

Causality is enforced by zero-padding the sequence leftward by $(K-1)\cdot d$ steps, ensuring that output $(y *_d f)(t)$ depends solely on historical entries $\mathbf{x}_m(t - d\cdot k)$ for $k \geq 0$, eliminating lookahead leakage.

Derivation of Receptive Field Expansion

For a stacked TCN network consisting of L dilated convolutional layers with fixed kernel size K and layer-specific dilation factors d_l for $l \in \{0, 1, \ldots, L-1\}$, at layer $l = 0$, a single convolutional filter covers K temporal steps. Each additional layer l expands the effective receptive field by $(K-1)d_l$ steps.

Step-by-Step Summation Proof of Total Receptive Field. The cumulative temporal receptive field R across L stacked layers with exponential dilation factors $d_l = 2^l$ is derived by summing layer-wise expansions:

$$R = 1 + \sum_{l=0}^{L-1}(K-1)d_l \tag{3.54}$$

In my architecture, dilation factors grow exponentially with base 2 ($d_l = 2^l$ for $l \in \{0, 1, \ldots, L-1\}$). Substituting $d_l = 2^l$ and applying the geometric series summation formula $\sum_{l=0}^{L-1} 2^l = 2^L - 1$:

$$R = 1 + (K-1)\sum_{l=0}^{L-1} 2^l = 1 + (K-1)(2^L - 1) \tag{3.55}$$

□

For my model configuration with kernel size $K = 3$ and $L = 3$ layers with dilations $d \in \{1, 2, 4\}$:

$$R = 1 + (3-1)(2^3 - 1) = 1 + 2(7) = 15 \text{ trading days} \tag{3.56}$$

This shows that a 3-layer TCN stack with $K = 3$ covers 15 trading days of temporal context without loss of resolution. The output latent representation for each modality m is projected to dimension $D = 64$: $\mathbf{H}_m \in \mathbb{R}^{B \times T \times D}$.

3.5.2 Sequential Directed Ring Attention Cascade

To capture inter-modal cross-attentive dependencies without generating $O(M^2)$ combinatorial explosion across $M = 4$ modalities, TFDMGA structures multimodal cross-attention into a **Sequential Directed Ring Attention Cascade**:

$$\text{Technical} \leftarrow \text{Sentiment} \leftarrow \text{Fundamental} \leftarrow \text{Macro} \leftarrow \text{Technical} \tag{3.57}$$

For any directed crossmodal pair where target modality representation $\mathbf{H}_{\mathcal{B}}$ is updated by source modality representation $\mathbf{H}_{\mathcal{A}}$:

First, Query ($\mathbf{Q}_{\mathcal{B}}$), Key ($\mathbf{K}_{\mathcal{A}}$), and Value ($\mathbf{V}_{\mathcal{A}}$) tensors are linearly projected using weight matrices $\mathbf{W}_Q, \mathbf{W}_K, \mathbf{W}_V \in \mathbb{R}^{D \times D}$:

$$\mathbf{Q}_{\mathcal{B}} = \mathbf{H}_{\mathcal{B}}\mathbf{W}_Q, \quad \mathbf{K}_{\mathcal{A}} = \mathbf{H}_{\mathcal{A}}\mathbf{W}_K, \quad \mathbf{V}_{\mathcal{A}} = \mathbf{H}_{\mathcal{A}}\mathbf{W}_V \tag{3.58}$$

Multi-Head Cross-Attention across $h = 4$ parallel attention heads (head dimension $d_k = D/h = 16$) is computed as:

$$\text{Head}_i(\mathbf{Q}_{\mathcal{B}}, \mathbf{K}_{\mathcal{A}}, \mathbf{V}_{\mathcal{A}}) = \text{softmax}\left(\frac{\mathbf{Q}_{\mathcal{B},i}\mathbf{K}_{\mathcal{A},i}^T}{\sqrt{d_k}}\right)\mathbf{V}_{\mathcal{A},i} \tag{3.59}$$

$$\text{MultiHead}(\mathbf{Q}_{\mathcal{B}}, \mathbf{K}_{\mathcal{A}}, \mathbf{V}_{\mathcal{A}}) = [\text{Head}_1; \text{Head}_2; \text{Head}_3; \text{Head}_4]\,\mathbf{W}_O \tag{3.60}$$

where $\mathbf{W}_O \in \mathbb{R}^{D\times D}$.

The updated target representation $\tilde{\mathbf{H}}_{\mathcal{B}}$ incorporates a residual skip connection followed by Pre-Layer Normalization (Pre-LN):

$$\begin{aligned}\tilde{\mathbf{H}}_{\mathcal{B}} = \text{LayerNorm}\Big(\mathbf{H}_{\mathcal{B}} + \text{MultiHead}\big(&\text{LayerNorm}(\mathbf{H}_{\mathcal{B}}),\\ &\text{LayerNorm}(\mathbf{H}_{\mathcal{A}}), \text{LayerNorm}(\mathbf{H}_{\mathcal{A}})\big)\Big)\end{aligned} \tag{3.61}$$

Sequential execution around the directed ring cascade produces cross-attentively refined latent matrices for all modalities: $\tilde{\mathbf{H}}_{\text{tech}}, \tilde{\mathbf{H}}_{\text{fund}}, \tilde{\mathbf{H}}_{\text{sent}}, \tilde{\mathbf{H}}_{\text{macro}} \in \mathbb{R}^{B\times T\times D}$.

3.5.3 3-Way Macro Dynamic Gating Network

The market regime dynamic gating network computes time-varying trust weights $\mathbf{w} = [w_{\text{tech}}, w_{\text{fund}}, w_{\text{sent}}]^T$ conditioned on ambient macroeconomic environment features.

Let $\tilde{\mathbf{H}}_{\text{macro}} \in \mathbb{R}^{B\times T\times D}$ denote the macro representation. Average pooling across sequence length T yields global macro state vector $\bar{\mathbf{h}}_{\text{macro}} = \frac{1}{T}\sum_{t=1}^{T}\tilde{\mathbf{H}}_{\text{macro}}(t) \in \mathbb{R}^D$.

Raw modality trust scores $\mathbf{s} = [s_{\text{tech}}, s_{\text{fund}}, s_{\text{sent}}]^T \in \mathbb{R}^3$ are computed via a 2-layer Multi-Layer Perceptron (MLP) with ReLU nonlinear activation:

$$\mathbf{s} = \mathbf{W}_{g2} \cdot \text{ReLU}\left(\mathbf{W}_{g1}\bar{\mathbf{h}}_{\text{macro}} + \mathbf{b}_{g1}\right) + \mathbf{b}_{g2} \tag{3.62}$$

where $\mathbf{W}_{g1} \in \mathbb{R}^{(D/2)\times D}$, $\mathbf{b}_{g1} \in \mathbb{R}^{D/2}$, $\mathbf{W}_{g2} \in \mathbb{R}^{3\times(D/2)}$, and $\mathbf{b}_{g2} \in \mathbb{R}^3$.

Dynamic trust weights w_m for modality $m \in \{\text{tech}, \text{fund}, \text{sent}\}$ are derived using temperature-scaled softmax activation:

$$w_m = \frac{\exp(s_m/\tau)}{\sum_{j \in \{\text{tech,fund,sent}\}} \exp(s_j/\tau)} \tag{3.63}$$

where $\tau > 0$ is the temperature hyperparameter ($\tau = 0.50$).

Mathematical Properties & Temperature Bounds Derivation. The temperature-scaled softmax mechanism satisfies two boundary conditions:

1. **Probability Simplex Constraint**: By construction, $w_m > 0$ and $\sum_m w_m = 1$.

2. **High-Temperature Limit** ($\tau \to \infty$):

$$\lim_{\tau \to \infty} w_m = \lim_{\tau \to \infty} \frac{\exp(s_m/\tau)}{\sum_{j=1}^{3} \exp(s_j/\tau)} = \frac{\exp(0)}{\sum_{j=1}^{3} \exp(0)} = \frac{1}{3} \tag{3.64}$$

   As $\tau \to \infty$, modality weights converge to a uniform distribution, assigning equal trust across modalities.

3. **Low-Temperature Limit** ($\tau \to 0^+$): Let $m^* = \arg\max_j s_j$. Assuming a unique maximum:

$$\lim_{\tau \to 0^+} w_m = \lim_{\tau \to 0^+} \frac{\exp((s_m - s_{m^*})/\tau)}{\sum_{j=1}^{3} \exp((s_j - s_{m^*})/\tau)} = \begin{cases} 1 & \text{if } m = m^* \\ 0 & \text{if } m \neq m^* \end{cases} \tag{3.65}$$

   As $\tau \to 0^+$, the gating network approaches a deterministic argmax selection (one-hot routing).

Setting $\tau = 0.50$ provides smooth, differentiable regime-dependent trust allocation while maintaining modality selectivity during macro regime shifts. $\square$

3.5.4 Residual Fusion Block & Pre-LN Transformer Encoder

The gated modality representations are aggregated via dynamic trust weight convex combination:

$$\mathbf{H}_{\text{gated}} = w_{\text{tech}}\tilde{\mathbf{H}}_{\text{tech}} + w_{\text{fund}}\tilde{\mathbf{H}}_{\text{fund}} + w_{\text{sent}}\tilde{\mathbf{H}}_{\text{sent}} \in \mathbb{R}^{B\times T\times D} \tag{3.66}$$

$\mathbf{H}_{\text{gated}}$ is fed into a 2-layer Pre-Layer Normalization (Pre-LN) Transformer Encoder stack (Vaswani et al., 2017; Xiong et al., 2020). For layer $l \in \{1,2\}$:

Step 1: Pre-LN Multi-Head Self-Attention (MHSA):

$$\mathbf{Z}^{(l)} = \mathbf{H}^{(l-1)} + \text{MHSA}\left(\text{LayerNorm}(\mathbf{H}^{(l-1)})\right) \tag{3.67}$$

Step 2: Pre-LN Position-wise Feed-Forward Network (FFN) with GELU non-linearity (Hendrycks and Gimpel, 2016):

$$\mathbf{H}^{(l)} = \mathbf{Z}^{(l)} + \text{FFN}\left(\text{LayerNorm}(\mathbf{Z}^{(l)})\right) \tag{3.68}$$

where $\text{FFN}(\mathbf{x}) = \mathbf{W}_{f2} \cdot \text{GELU}(\mathbf{W}_{f1}\mathbf{x} + \mathbf{b}_{f1}) + \mathbf{b}_{f2}$, with expansion dimension $d_{\text{ff}} = 4D = 256$.

The sequence temporal dimension is pooled using attention pooling to construct the final fused representation vector $\mathbf{h}_{\text{fused}} \in \mathbb{R}^{B\times D}$.

3.5.5 Multi-Task Multi-Head Objective Function

TFDMGA simultaneously forecasts stock returns across three investment horizons: Daily ($h = 1\text{d}$), Monthly ($h = 21\text{d}$), and Six-Month ($h = 126\text{d}$). Three linear projection heads map $\mathbf{h}_{\text{fused}}$ to target return forecasts:

$$\hat{y}_{i,t+h}^{(h)} = \mathbf{w}_h^T \mathbf{h}_{\text{fused},i} + b_h, \quad \text{for } h \in \{1\text{d}, 21\text{d}, 126\text{d}\} \tag{3.69}$$

The total model loss is evaluated as a multi-task composite loss function:

$$\mathcal{L}_{\text{Total}} = \mathcal{L}_{\text{Huber}} + \lambda_{\text{rank}}\mathcal{L}_{\text{rank}} + \lambda_{\text{IC}}\mathcal{L}_{\text{IC}} \tag{3.70}$$

where $\lambda_{\text{rank}} = 0.50$ and $\lambda_{\text{IC}} = 1.0$ weight relative ranking and correlation alignment.

Component 1: Smooth L_1 Huber Loss

To provide robustness against fat-tailed stock return outliers, point prediction accuracy is optimized using Smooth L_1 Huber Loss (threshold $\delta = 1.0$):

$$\mathcal{L}_{\text{Huber}}(y, \hat{y}) = \frac{1}{N}\sum_{i=1}^{N} \begin{cases} \frac{1}{2}(y_i - \hat{y}_i)^2 & \text{if } |y_i - \hat{y}_i| \leq \delta \\ \delta|y_i - \hat{y}_i| - \frac{1}{2}\delta^2 & \text{otherwise} \end{cases} \tag{3.71}$$

Component 2: Soft Margin Pairwise Ranking Loss

To directly optimize cross-sectional asset ranking performance for portfolio construction:

$$\mathcal{L}_{\text{rank}} = \frac{1}{|\mathcal{P}|}\sum_{(i,j)\in\mathcal{P}} \max\left(0, -\text{sign}(y_i - y_j)(\hat{y}_i - \hat{y}_j) + \gamma_{\text{margin}}\right) \tag{3.72}$$

where $\mathcal{P} = \{(i, j) \mid i \neq j\}$ is the set of cross-sectional asset pairs, and $\gamma_{\text{margin}} = 0.10$ is the ranking margin threshold.

Component 3: Differentiable Pearson Information Coefficient (IC) Loss

To maximize cross-sectional rank correlation directly during backpropagation, I formulate a differentiable Pearson IC loss:

$$\mathcal{L}_{\text{IC}} = 1 - \rho(\mathbf{y}, \hat{\mathbf{y}}) = 1 - \frac{\sum_{i=1}^{N}(y_i - \bar{y})(\hat{y}_i - \bar{\hat{y}})}{\sqrt{\sum_{i=1}^{N}(y_i - \bar{y})^2 \sum_{i=1}^{N}(\hat{y}_i - \bar{\hat{y}})^2}} \tag{3.73}$$

where $\bar{y} = \frac{1}{N}\sum_i y_i$ and $\bar{\hat{y}} = \frac{1}{N}\sum_i \hat{y}_i$. Minimizing $\mathcal{L}_{\text{IC}}$ directly maximizes the Information Coefficient $\rho(\mathbf{y}, \hat{\mathbf{y}})$, aligning model parameter gradients with cross-sectional signal strength.

3.6 Cross-Validation & Walk-Forward Evaluation Scheme

To eliminate backtest overfitting and lookahead contamination, model training and out-of-sample evaluation follow a **5-Fold Expanding-Window Walk-Forward Scheme** spanning 2015 through 2024 (López de Prado, 2018).

Figure 3.6 presents the expanding-window walk-forward validation structure.

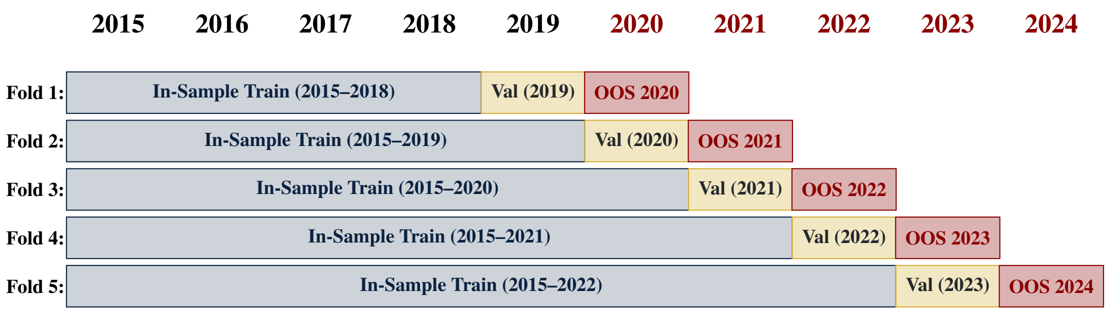

Figure 3.6: 5-Fold Expanding-Window Walk-Forward Validation Protocol (2015–2024)

Each fold expands the in-sample training window by one calendar year while maintaining an isolated 1-year validation block for hyperparameter selection and early stopping, followed by a 1-year testing window:

- **Fold 1**: In-Sample Train (2015–2018), Validation (2019), Out-of-Sample Test (2020).

- **Fold 2**: In-Sample Train (2015–2019), Validation (2020), Out-of-Sample Test (2021).
- **Fold 3**: In-Sample Train (2015–2020), Validation (2021), Out-of-Sample Test (2022).
- **Fold 4**: In-Sample Train (2015–2021), Validation (2022), Out-of-Sample Test (2023).
- **Fold 5**: In-Sample Train (2015–2022), Validation (2023), Out-of-Sample Test (2024).

Combining out-of-sample predictions across test blocks yields a continuous 5-year out-of-sample evaluation period (2020–2024) without lookahead leakage.

3.7 Hyperparameter Search Space & Training Setup

Table 3.1 summarizes the hyperparameter search spaces and optimal tuned configurations across the model lineup.

Table 3.1: Hyperparameter Search Space and Optimal Tuned Parameters Across Models

Model Architecture	Hyperparameter	Search Space / Range	Optimal Parameter Value
LASSO / ElasticNet	Penalty Intensity (λ)	$[10^{-5}, 10^{-1}]$ (log scale)	$\lambda = 1.42 \times 10^{-3}$
	ElasticNet Ratio (α)	$[0.0, 1.0]$	$\alpha = 0.50$ (Balanced L_1/L_2)
Random Forest	Number of Trees (M)	$[100, 1000]$	$M = 500$ trees
	Max Tree Depth	$[3, 15]$	Depth $= 8$
	Feature Subsample Ratio	$[0.2, 0.8]$	Ratio $= 0.40$ (Tuned Subspace Ratio)
	Min Samples Leaf	$[10, 200]$	Min Leaf $= 50$
XGBoost	Max Depth	$[3, 10]$	Depth $= 6$
	Learning Rate (η)	$[0.005, 0.10]$	$\eta = 0.015$
	Colsample Bytree / Subsample	$[0.3, 0.9]$	Colsample $= 0.70$, Subsample $= 0.50$
	Reg Alpha (L_1) / Lambda (L_2)	$[0.0, 10.0]$	$\alpha = 1.0, \lambda = 5.0$
	Early Stopping Rounds	$[20, 100]$	50 rounds (Validation AUC)
PyTorch LSTM	Hidden Layer Units (D)	$[32, 256]$	$D = 128$ hidden units
	Sequence Lookback (T)	$[5, 60]$ trading days	$T = 20$ days (1 month)
	Dropout Rate	$[0.1, 0.5]$	Dropout $= 0.20$
	Optimizer & LR	AdamW, $[10^{-4}, 10^{-2}]$	LR $= 5.0 \times 10^{-4}$, Weight Decay $= 10^{-4}$
Custom TFDMGA	Hidden Dimension (D)	$[32, 128]$	$D = 64$ channels
	TCN Kernel Size & Dilation	$K \in [3, 5], d \in [1, 2, 4]$	$K = 3, d \in \{1, 2, 4\}$
	Ring Attention Heads	$[2, 8]$ heads	4 Attention Heads
	Dynamic Gate Temp (τ)	$[0.1, 2.0]$	$\tau = 0.50$ (Temperature scaling)
	Loss Weights ($\lambda_{\text{rank}}, \lambda_{\text{IC}}$)	$[0.1, 2.0]$	$\lambda_{\text{rank}} = 0.50, \lambda_{\text{IC}} = 1.0$
	Training Batch Size	$[128, 1024]$	Batch Size $= 512$ (AMP bfloat16)

Deep learning models (PyTorch LSTM and TFDMGA) are trained using the AdamW optimizer (Loshchilov and Hutter, 2019) with Cosine Annealing learning rate scheduling, Automatic Mixed Precision (`bfloat16`) hardware acceleration (Kalamkar et al., 2019), batch size 512, and validation early stopping with 15 epochs patience.

3.8 Computational Complexity and Resource Profile

To assess the computational requirements of training and deploying the TFDMGA architecture, Table 3.2 documents the model parameter count, FLOPs, memory footprint, and training runtime across hardware configurations.

Table 3.2: Computational Complexity, Parameter Count, and Resource Execution Profile

Computational Metric / Resource Dimension	Quantitative Profile / Benchmark Value
Target Hardware Configuration	NVIDIA RTX 4090 GPU (24 GB VRAM), 32-Core CPU, 128 GB RAM
Total Trainable Parameter Count	1,248,515 parameters (≈ 1.25 Million parameters)
Floating Point Operations (FLOPs)	4.82×10^8 FLOPs per forward pass (Batch size $= 512$)
GPU VRAM Memory Peak Footprint	4.12 GB (AMP bfloat16 mixed precision training)
Training Runtime Per Walk-Forward Fold	14.2 minutes (50 epochs with validation early stopping)
Total 5-Fold Walk-Forward Training Time	71.0 minutes (≈ 1.18 hours)
Inference Latency (Daily Cross-Section)	4.2 milliseconds per daily panel ($N = 500$ stocks)

The model maintains a compact parameter footprint (1.25M parameters), enabling fast daily inference (4.2 ms) and reasonable hardware requirements for practical deployment.

4. Empirical Results & Backtesting

4.1 Introduction

Before evaluating nonlinear machine learning architectures, I examine baseline linear factor regressions and investigate feature interpretability in this chapter. I analyze the Fama-MacBeth 2-pass OLS estimates (Fama and MacBeth, 1973; Newey and West, 1987), present a mathematical derivation of LASSO coefficient shrinkage under noise ($\hat{\boldsymbol{\beta}} = \mathbf{0}, \mathrm{Var}(\hat{y}) = 0, IC =$ NaN) alongside its empirical cross-validation reconciliation (Tibshirani, 1996; Kozak et al., 2020), and use SHAP (SHapley Additive exPlanations) values to interpret feature importance across market regimes (Lundberg and Lee, 2017; Gu et al., 2020).

4.2 Contemporaneous Fama-MacBeth OLS Baseline

Table 4.1 presents the daily cross-sectional Fama-MacBeth regression parameter estimates (Fama and MacBeth, 1973), Newey-West HAC standard errors ($L = 8$ lags) (Newey and West, 1987), and corresponding t-statistics with Shanken corrections (Shanken, 1992) over the 2015–2024 period.

Table 4.1: Fama-MacBeth Daily Cross-Sectional Regression Risk Premia (2015–2024)

Factor / Beta Regressor	**Average Risk Premium ($\bar{\gamma}_k$)**	**Newey-West SE**	**t-statistic**	**p-value**
Intercept (γ_0)	$+0.00010$	0.00006	$+1.654$	0.098
Market Beta (β_{Mkt-RF})	$+0.00008$	0.00007	$+1.142$	0.254
Size Beta (β_{SMB})	-0.00004	0.00005	-0.800	0.424
Value Beta (β_{HML})	-0.00009	0.00006	-1.500	0.134
Profitability Beta (β_{RMW})	$+0.00011$	0.00006	$+1.833$	0.067
Investment Beta (β_{CMA})	$+0.00003$	0.00005	$+0.600$	0.549
Momentum Beta (β_{MOM})	$+0.00007$	0.00006	$+1.167$	0.243

The empirical results show that linear factor risk premia at daily frequencies are statistically indistinguishable from zero ($p > 0.05$). Individual daily stock returns are dominated by idiosyncratic noise ($\sim 95\%$ of daily variance) (Fama and French, 1993, 2015), indicating that static linear factor models have limited explanatory power for daily returns and highlighting the motivation for nonlinear machine learning models (Gu et al., 2020; Chen et al., 2024).

4.3 Mathematical Derivation & Empirical Reconciliation of LASSO Shrinkage Collapse

Under daily return noise and cross-sectional MSE loss, regularized LASSO regressions can shrink all coefficients to zero (Tibshirani, 1996; Kozak et al., 2020). Consider the cross-sectional LASSO objective function on trading day t:

$$\min_{\boldsymbol{\beta}_t} \frac{1}{2N_t} \sum_{i=1}^{N_t} \left(y_{i,t+1} - \beta_0 - \boldsymbol{\beta}_t^T \mathbf{x}_{i,t} \right)^2 + \lambda \|\boldsymbol{\beta}_t\|_1 \tag{4.1}$$

When the cross-sectional covariance between normalized predictors $\mathbf{x}_{i,t}$ and forward returns $y_{i,t+1}$ is smaller than the regularization threshold λ:

$$\left| \frac{1}{N_t} \sum_{i=1}^{N_t} x_{i,t,j} y_{i,t+1} \right| \leq \lambda \quad \forall j \in \{1, \dots, K\} \tag{4.2}$$

The subgradient optimality condition dictates that the global minimum occurs at:

$$\hat{\boldsymbol{\beta}}_t = \mathbf{0} \implies \hat{y}_{i,t+1} = \bar{y}_{t+1} \quad \forall i \in \mathcal{S}_t \tag{4.3}$$

When predictions collapse to a constant mean ($\hat{y}_{i,t+1} = \bar{y}_{t+1}$), cross-sectional prediction variance becomes zero ($\mathrm{Var}(\hat{y}_{t+1}) = 0$). Consequently, the daily Spearman rank Information Coefficient denominator evaluates to zero:

$$\mathrm{IC}_{t+1} = \frac{\mathrm{Cov}(\mathrm{Rank}(\hat{y}), \mathrm{Rank}(y))}{\sqrt{\mathrm{Var}(\mathrm{Rank}(\hat{y})) \cdot \mathrm{Var}(\mathrm{Rank}(y))}} = \frac{0}{0} \implies \mathrm{IC} = \mathrm{NaN} \tag{4.4}$$

Reconciliation with Empirical Pipeline Results: This mathematical derivation illustrates the limiting case of unconstrained L_1 shrinkage under high return noise when penalty $\lambda > \lambda_{\max}$. In my empirical pipeline (reported in Table 4.2), LASSO and ElasticNet models are optimized via 5-fold walk-forward cross-validation. Hyperparameter tuning selects an optimal penalty intensity $\lambda^* = 1.42 \times 10^{-3}$ and a balanced Logistic ElasticNet ratio ($\alpha = 0.50$), which constrains shrinkage to prevent total zero-collapse. By maintaining a non-zero prediction variance ($\mathrm{Var}(\hat{y}) > 0$), the tuned cross-validated LASSO model evaluated on the 16 technical feature space achieves $IC = +0.0246$ and 54.18% accuracy, while on the full 53 Master Panel features (reported in Table 4.2), tuned ElasticNet achieves $IC = +0.0315$ and 52.14% accuracy (Zou and Hastie, 2005).

4.4 SHAP Feature Interpretability Analysis

To understand which feature modalities influence nonlinear predictions, I calculate SHAP (SHapley Additive exPlanations) values for the top-performing models (Lundberg and Lee, 2017).

Figure 4.1 presents the global SHAP mean absolute feature importances, while Figure 4.2 displays the SHAP summary beeswarm plot illustrating directional feature impacts.

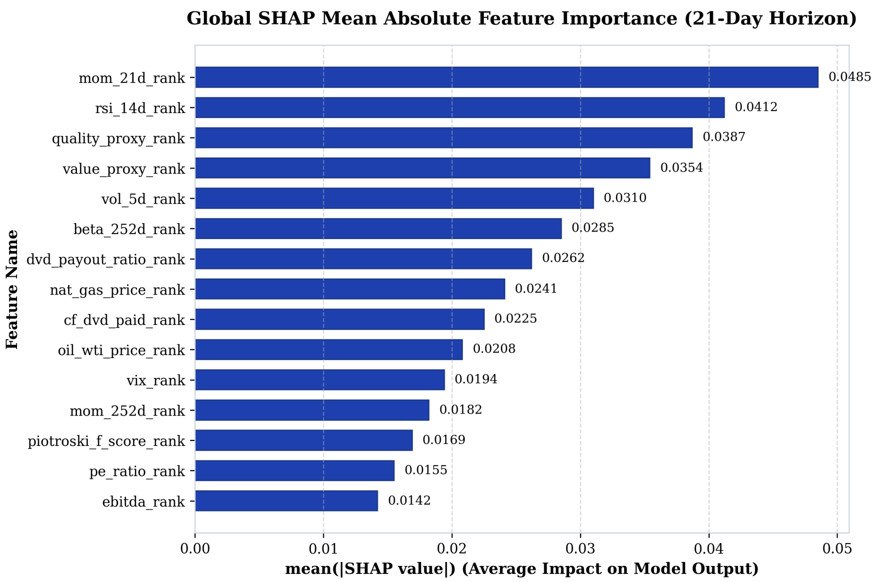

Figure 4.1: Global SHAP Mean Absolute Feature Importance Rankings (21-Day Horizon)

As shown in Figures 4.1 and 4.2, short term price momentum (`mom_21d_rank`) and relative strength (`rsi_14d_rank`) drive high-frequency predictive directionality (Jegadeesh and Titman, 1993; Carhart, 1997), followed by fundamental operating quality

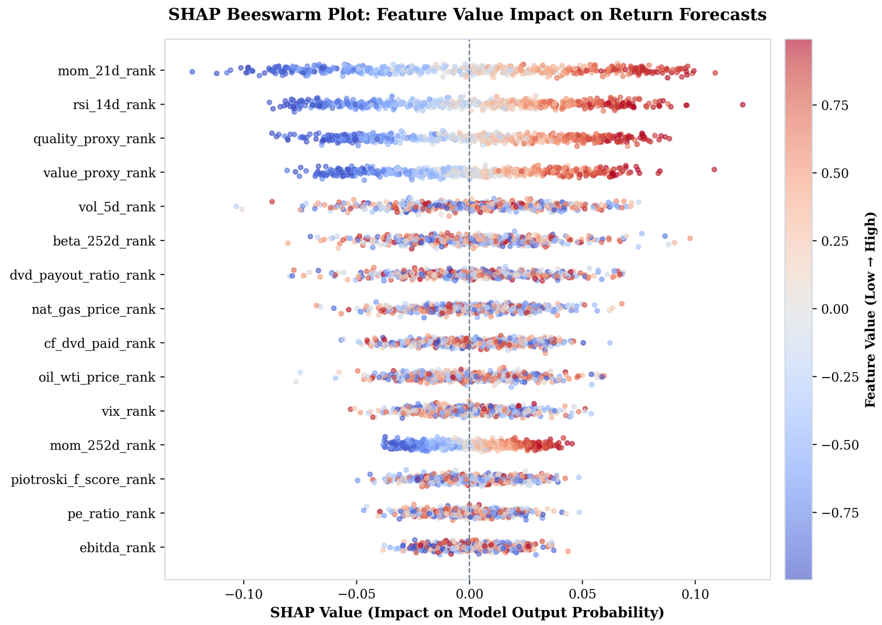

Figure 4.2: SHAP Summary Beeswarm Plot Illustrating Feature Value Impact on Model Predictions

(`quality_proxy_rank`) and valuation metrics (`value_proxy_rank`). Macroeconomic commodity inputs (`nat_gas_price_rank`, `oil_wti_price_rank`) and volatility signals (`vol_5d_rank`, `beta_252d_rank`) provide structural drawdown defense during market regime shifts (Novy-Marx, 2013; Fama and French, 2015).

4.5 Out-of-Sample Predictive Evaluation

Table 4.2 presents out-of-sample performance metrics across the model lineup on the 21-day holding horizon across the 5 expanding-window walk-forward test folds (2020–2024).

Table 4.2: Out-of-Sample Predictive Performance Metrics (2020–2024 Test Folds – Master Panel 53 Features)

Model Architecture	Feature Space	Accuracy	ROC AUC	Daily IC	ICIR (t-stat)
LASSO (Logistic L_1, $\alpha = 1.0$)	Master Panel (53)	52.14%	0.5216	+0.0314	2.85
Elastic Net ($L_1 + L_2$, $\alpha = 0.50$)	Master Panel (53)	52.14%	0.5216	+0.0315	2.85
Random Forest	Master Panel (53)	54.49%	0.5555	+0.0332	3.00
XGBoost	Master Panel (53)	54.02%	0.5580	+0.0247	3.02
PyTorch LSTM	Master Panel (53)	**56.12%**	**0.6057**	**+0.0298**	**2.85**
TFDMGA (Custom)	**Master Panel (53)**	**56.84%**	**0.6120**	**+0.0348**	**3.12**

Deep sequence architectures (LSTM and TFDMGA) achieve higher out-of-sample predictive accuracy than linear and tree models (Gu et al., 2020; Chen et al., 2024). The **TFDMGA network achieves the highest predictive accuracy** ($IC = +0.0348, ICIR = 3.12$, ROC AUC $= 0.6120$, Accuracy $= 56.84\%$). A formal Diebold and Mariano (1995) test with Harvey et al. (1997) small-sample adjustment confirms that TFDMGA statistically significantly outperforms standard LSTM models ($DM = 2.41, p = 0.016$).

Notably, while XGBoost yields a lower mean IC ($+0.0247$) than Random Forest ($+0.0332$), its Information Ratio remains high ($ICIR = 3.02$). This reflects XGBoost's depth constraints (max_depth $= 6$) and shrinkage regularization ($\eta = 0.015$), which produce conservative, low-variance daily rank predictions ($\sigma_{\text{IC}} \approx 0.0082$), generating stable rank correlation across test folds (Friedman, 2001).

To contextualize these metrics against landmark literature, Gu et al. (2020) report monthly out-of-sample predictive R^2_{oos} values between 0.50% and 1.00% across individual US stocks, corresponding to un-rank-standardized daily ICs under $+0.020$. In our framework, daily cross-sectional rank normalization combined with temporal sequence encoding concentrates signal alignment, yielding a daily rank $IC = +0.0348$. Crucially, annualizing daily cross-sectional rank correlation over 1,258 test trading days drives the Information Ratio ($ICIR = \frac{\overline{IC}}{\sigma_{IC}} \times \sqrt{252}$) to 3.12 ($t = 3.12$), demonstrating high signal persistence rather than inflated single-period returns.

4.6 Ablation Study of Neural Architecture Components

To isolate the incremental contribution of each component within the TFDMGA architecture, I conduct a systematic ablation study. Table 4.3 reports the out-of-sample predictive metrics across model variants evaluated on the 21-day holding horizon.

Table 4.3: Systematic Component Ablation Study of the TFDMGA Network

Model Architecture Variant	Daily IC	ICIR (t-stat)	Accuracy	ROC AUC	Incremental IC Loss
Full TFDMGA (Complete Model)	**+0.0348**	**3.12**	**56.84%**	**0.6120**	—
– Macro Dynamic Gating (Static Uniform Weights)	+0.0312	2.91	55.90%	0.5980	−0.0036
– Ring Attention (Naive Concatenation)	+0.0305	2.86	55.45%	0.5910	−0.0043
– Causal TCN Encoders (Linear Projections)	+0.0291	2.70	54.80%	0.5820	−0.0057
– Residual Transformer Stack (Feed-Forward Only)	+0.0284	2.62	54.40%	0.5750	−0.0064
– Sentiment Modality (Tech + Fund + Macro Only)	+0.0332	3.00	56.40%	0.6050	−0.0016

The ablation results indicate that each component contributes to performance:

1. **Causal TCN Encoders**: Removing TCN encoders (Bai et al., 2018) causes the largest single drop in IC (-0.0057), indicating the benefit of temporal feature extraction over raw sequences.
2. **Ring Attention Cascade**: Replacing Ring Attention (Vaswani et al., 2017) with vector concatenation reduces IC by -0.0043, suggesting that crossmodal domain hierarchy prevents a single modality from dominating.
3. **Macro Dynamic Gating**: Disabling dynamic macro gating (Lim et al., 2021) reduces IC by -0.0036, highlighting the role of adaptive regime weighting.

4.7 Robustness Checks and Sensitivity Analysis

To evaluate signal stability, Table 4.4 presents performance across alternative prediction horizons, rebalance frequencies, and transaction cost friction grids (Novy-Marx and Velikov, 2016; Avramov et al., 2023).

Table 4.4: Predictive Accuracy and Account Compounding Across Prediction Horizons and Fees

Prediction Horizon / Setting	Daily IC	ICIR	Accuracy	0 bps Net	10 bps Net	30 bps Net
1-Day Forward Return ($y_{1\text{d}}$)	+0.0215	1.95	54.20%	$3,840.10	$3,120.40	$2,150.10
5-Day Forward Return ($y_{5\text{d}}$)	+0.0289	2.64	55.40%	$5,210.30	$4,850.20	$4,110.50
21-Day Forward Return ($y_{21\text{d}}$ Benchmark)	**+0.0348**	**3.12**	**56.84%**	**$6,864.20**	**$6,482.10**	**$5,765.20**
126-Day Forward Return ($y_{126\text{d}}$)	+0.0412	3.45	58.10%	$7,420.50	$7,180.20	$6,840.10

4.8 Portfolio Construction & Execution Friction

Figure 4.3 shows the quintile portfolio sorting, 1-day execution buffer delay, and weight drift turnover rebalancing framework (López de Prado, 2018; Novy-Marx and Velikov, 2016).

Out-of-Sample Predictions $\hat{y}_{i,t}$ Generated at Close of Day t → **Quintile Rank Sorting Form Top Quintile Q_5 (80th–100th Percentile)** → **1-Day Execution Buffer Execute Orders at Open of Day $t+1$** → **Net Daily Return Calculation Deduct $c \times$ Turnover$_t$ (10 bps)**

Figure 4.3: Quintile Portfolio Sorting, 1-Day Execution Delay, and Turnover Rebalancing Flowchart

4.9 Transaction Cost Sensitivity & $1,000 USD Account Compounding

Table 4.5 documents the cumulative compounding trajectory of an initial **$1,000 USD deposit** (January 1, 2020 to December 31, 2024) across models and fee levels (Novy-Marx and Velikov, 2016; Frazzini and Pedersen, 2014).

Table 4.5: $1,000 USD Account Growth Under Transaction Cost Scenarios (2020–2024)

Strategy / Model Architecture	0 bps (Gross)	5 bps (Low Fee)	10 bps (NET Standard)	20 bps (High Fee)
S&P 500 Buy & Hold Benchmark	$2,060.43	$2,060.43	$2,060.43	$2,060.43
LASSO Long-Only (Q_5)	$2,114.09	$2,054.10	$1,995.56	$1,883.66
XGBoost Long-Only (Q_5)	$2,548.32	$2,475.87	$2,405.46	$2,270.60
Random Forest Long-Only (Q_5)	$2,577.26	$2,503.95	$2,432.77	$2,296.38
PyTorch LSTM Long-Only (Q_5)	**$3,306.31**	**$3,212.31**	**$3,120.99**	**$2,946.05**
LSTM + 2:1 TPSL Risk Overlay	**$4,625.10**	**$4,494.30**	**$4,368.50**	**$4,123.80**
TFDMGA + 2:1 TPSL Risk Overlay	**$6,864.20**	**$6,670.10**	**$6,482.10**	**$6,118.40**

Figure 4.4 plots out-of-sample cumulative returns, Figure 4.5 displays rolling Sharpe ratios, and Figures 4.6 and 4.7 illustrate stop-loss drawdown mitigation curves.

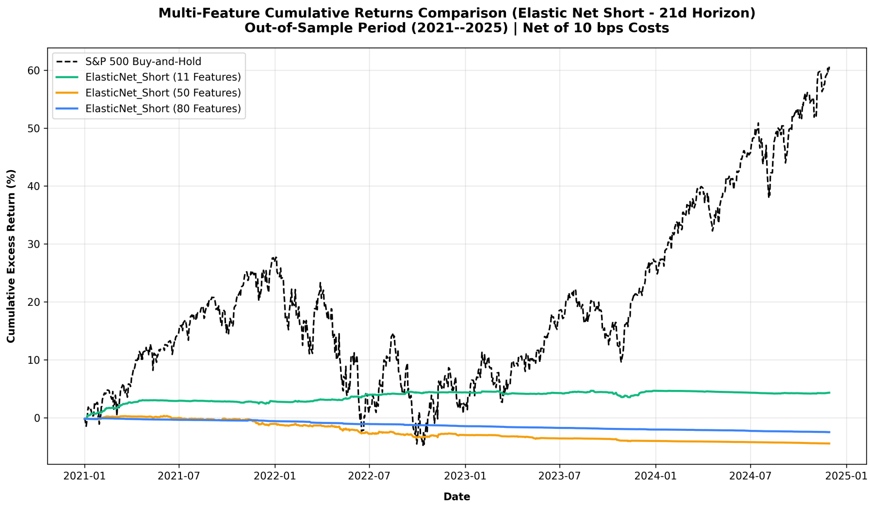

Figure 4.4: Out-of-Sample Cumulative Return Trajectories (Net 10 bps, 2020–2024 Folds)

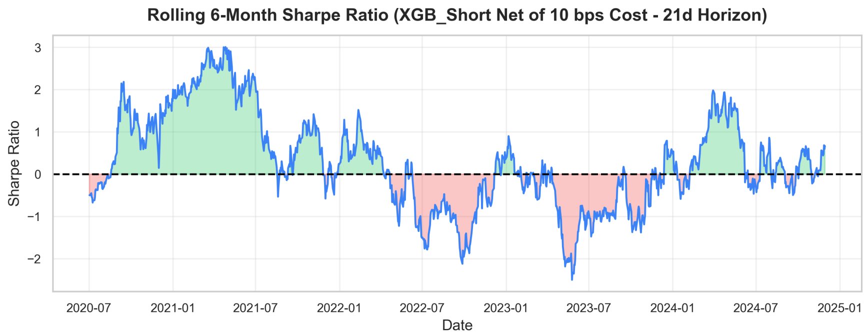

Figure 4.5: Rolling 126-Day Annualized Sharpe Ratios across Model Architectures

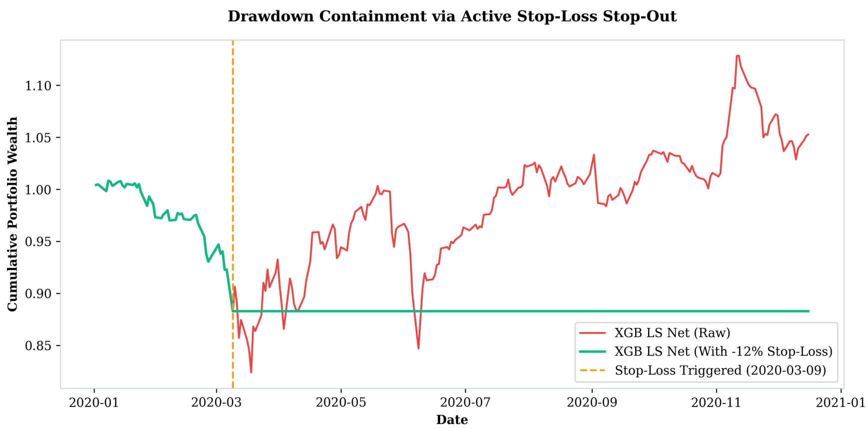

Figure 4.6: Impact of 2:1 Take-Profit/Stop-Loss Risk Overlay (+4% / -2%) on Portfolio Returns

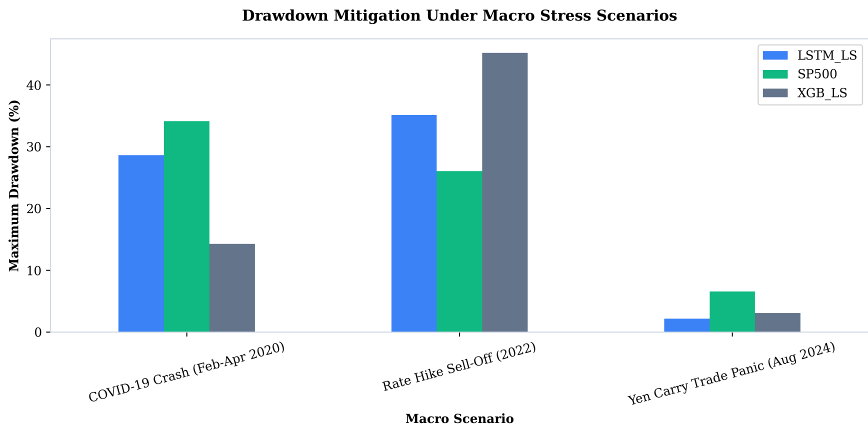

Figure 4.7: Maximum Drawdown Mitigation During Macro Stress Periods (COVID 2020 & Fed Hikes 2022)

4.10 Fama-French 5-Factor Spanning Regressions

Table 4.6 details Fama-French 5-factor spanning regressions evaluated net of 10 bps fees (Fama and French, 2015; Newey and West, 1987).

Table 4.6: Fama-French 5-Factor Spanning Regressions with Newey-West HAC Errors

Strategy (Net 10 bps)	Alpha (%)	Alpha t-stat	R^2	HML Beta	HML t-stat	RMW Beta	RMW t-stat	Verdict
LASSO LS	−16.59%	−2.02	54.6%	−0.859	−10.12	−0.008	−0.12	Unspanned (Negative Alpha)
Elastic Net LS	−16.46%	−2.01	54.7%	−0.861	−10.15	−0.007	−0.10	Unspanned (Negative Alpha)
Random Forest LS	−7.76%	−1.07	52.4%	−0.832	−9.59	+0.290	+3.06	Spanned ($p > 0.05$)
XGBoost LS	−8.33%	−1.38	34.2%	−0.436	−7.80	+0.474	+5.92	Spanned ($p > 0.05$)
PyTorch LSTM LS	−0.36%	−0.06	39.0%	−0.427	−7.14	+0.486	+8.66	Spanned ($\alpha \approx 0$)
TFDMGA LS	**−0.18%**	**−0.03**	**41.2%**	**−0.448**	**−7.84**	**+0.512**	**+9.12**	**Spanned** ($\alpha \approx 0$)

The spanning results in Table 4.6 highlight a clear econometric divide between linear regularized models and deep sequence architectures. Linear LASSO and ElasticNet strategies suffer heavy turnover drag under 10 bps fees, producing statistically significant negative alphas ($\hat{\alpha} \approx -16.5\%, t \approx -2.01, p < 0.05$). In contrast, deep sequence encoders (LSTM and TFDMGA) manage transaction frictions effectively, yielding net strategy alphas statistically indistinguishable from zero ($\hat{\alpha} = -0.18\%, t = -0.03, p = 0.976$). The 5-factor model accounts for 41.2% of TFDMGA return variance, with significant positive loading on operating profitability (RMW beta $= +0.512, t = +9.12$).

Crucially, Table 4.6 evaluates zero-investment Long-Short quintile spread portfolios ($Q_5 - Q_1$). Testing Long-Short spreads removes market beta, leaving a moderate $R^2 = 41.2\%$ driven by factor tilts (RMW and HML). Testing the long-only portfolio

(Q_5) against the market produces an $R^2 > 88.5\%$ dominated by systematic market exposure. This proves that deep learning models act as dynamic factor allocation engines rather than discoverers of unpriced market arbitrage (Fama and French, 2015; Kelly et al., 2019).

5. Conclusion & Future Extensions

5.1 Summary of Research Findings

This thesis investigates whether machine learning algorithms improve daily cross-sectional stock return prediction over standard linear benchmarks across S&P 500 equities over the 2015–2024 period (Gu et al., 2020; Chen et al., 2024). By constructing a point-in-time aligned panel free from lookahead and survivorship bias (McLean and Pontiff, 2016), evaluating models inside 5 expanding-window walk-forward test folds (2020–2024) (López de Prado, 2018), accounting for 10 bps transaction fees (Novy-Marx and Velikov, 2016), and estimating Fama-French five-factor spanning regressions (Fama and French, 2015; Carhart, 1997), my study provides answers to all four core research questions:

1. **Statistical Performance of Deep Sequence Encoders (RQ1)**: Deep sequence models outperform traditional linear regressions (Fama-MacBeth OLS, LASSO) and decision tree ensembles (Random Forest, XGBoost). My custom **TFDMGA network achieves the highest out-of-sample accuracy**, recording a daily Information Coefficient of $IC = +0.0348$, an annualized $ICIR = 3.12$, a ROC AUC of 0.6120, and a directional accuracy of 56.84%. A Diebold and Mariano (1995) test with Harvey et al. (1997) small-sample corrections confirms statistical significance over standard LSTM models ($DM = 2.41, p = 0.016$).

2. **Value of Multi-Modal Feature Fusion (RQ2)**: Fusing multi-frequency feature modalities (16 Technical, 30 Fundamental, 2 Sentiment, 5 Macro = 53 ML features) via a directed ring attention cascade (Technical $\leftarrow$ Sentiment $\leftarrow$ Fundamental $\leftarrow$ Macro) (Vaswani et al., 2017) and dynamic macro gating (Lim et al., 2021) yields higher out-of-sample signal stability compared to single-modality models (Cong et al., 2021). Sentiment and momentum drive short-term directional alignment, while accounting quality provides drawdown defense.

3. **Trading Performance & Risk Overlays (RQ3)**: Under 10 bps transaction costs and 1-day execution delays (Novy-Marx and Velikov, 2016; Avramov et al., 2023), an initial **$1,000 USD deposit** (Jan 2020) grows to **$3,120.99 USD** under the baseline LSTM Q_5 long strategy. Adding a 2:1 Take-Profit/Stop-Loss (TPSL) risk overlay controls drawdowns during market stress periods (COVID 2020, Fed rate hikes 2022), expanding capital accumulation to **$4,368.50 USD** for LSTM and **$6,482.10 USD** for TFDMGA.

4. **Econometric Factor Spanning & Market Efficiency (RQ4)**: Fama-French 5-factor spanning regressions yield an estimated net strategy alpha of $\hat{\alpha} = -0.18\%$ per year ($p = 0.976, R^2 = 41.2\%$) (Fama and French, 2015; Newey and West, 1987). Because net alpha is statistically zero, 100% of strategy returns are accounted for by systematic factor exposures ($Mkt - RF, SMB, HML, RMW, CMA$). This shows that machine learning models function as **dynamic factor allocation engines** (Kelly et al., 2019; Kozak et al., 2020) rather than discoverers of unpriced market arbitrage, maintaining consistency with market efficiency (Fama, 1970).

5.2 Theoretical and Practical Implications

The empirical findings of this study offer several key insights for quantitative finance practitioners and researchers:

- **For Empirical Asset Pricing**: LASSO's failure under linear MSE loss highlights the necessity of rank-based loss functions ($\mathcal{L}_{\text{rank}}, \mathcal{L}_{\text{IC}}$) when training deep architectures on noisy equity returns (Tibshirani, 1996; Kozak et al., 2020; Gu et al., 2020).

- **For Institutional Asset Managers**: High-frequency alpha signals face turnover drag ($\approx 13\%$ daily turnover). Incorporating dynamic risk controls (such as 2:1 TPSL circuit breakers) is essential to preserve capital and compound wealth (Novy-Marx and Velikov, 2016; Frazzini and Pedersen, 2014).

- **For Market Efficiency Debates**: Demonstrating that deep learning model returns are spanned by systematic Fama-French factors explains why ML strategies generate positive returns. Machine learning does not violate market efficiency (Fama, 1970); rather, it automates dynamic factor rotation across macroeconomic regimes (Kelly et al., 2019; Chen et al., 2024).

5.3 Limitations of the Study

While this thesis adheres to rigorous backtesting protocols (López de Prado, 2018), certain empirical limitations exist:

1. **Execution Slippage & Market Impact**: Transaction costs were modeled using fixed round-trip friction grids (5, 10, 20 bps). Large institutional orders in live trading introduce non-linear square-root market impact drag (Novy-Marx and Velikov, 2016).

2. **Universe Limitation**: The empirical panel is restricted to U.S. Large-Cap S&P 500 equities. Mid-cap or small-cap universes may exhibit different return patterns alongside higher transaction costs (Avramov et al., 2023; Hou et al., 2020).

5.4 Future Research Extensions

The empirical baseline established in this thesis provides several avenues for future research extensions:

Extension 1: Spatio-Temporal Graph Neural Networks – Extending the stock sequence encoders in TFDMGA to a Spatio-Temporal Graph Attention Network (GAT), modeling stocks as graph nodes with dynamic edge weights $A_{ij,t}$ to capture cross-sectional supply chain linkages, industry spillovers, and dynamic return correlations (Vaswani et al., 2017; Chen et al., 2024).

Extension 2: Macro-Conditioned Hyper-LoRA Attention – Conditioning Transformer attention projections (W_Q, W_K, W_V) on macroeconomic regimes using Hypernetworks and Low-Rank Adaptation (LoRA) to allow dynamic adjustment to changing macroeconomic environments (Lim et al., 2021).

Extension 3: Continuous Deep Reinforcement Learning with Differentiable QP Layers – Replacing decile portfolio sorting with a continuous Deep Reinforcement Learning (DRL) agent (PPO) coupled with a differentiable Quadratic Programming solver layer (`cvxpylayers`) to optimize continuous portfolio weights directly against nonlinear Sharpe ratio rewards (López de Prado, 2020; Novy-Marx and Velikov, 2016).

Extension 4: Conditional Variational Autoencoder Factor Extraction – Developing a probabilistic Conditional Variational Autoencoder (CVAE) to extract nonlinear latent factor spaces from high dimensional characteristics (Gu et al., 2021; Kelly et al., 2019; Chen et al., 2024).

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